

JOSEPH AARON MUSARA

AI-Powered Full-Stack Software Engineer

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PROFESSIONAL SUMMARY

Results-driven Software Engineer with profound expertise in full-stack development, AI-integrated systems, and scalable backend architecture. Proven ability to design and deploy intelligent systems, including AI-powered recruitment platforms featuring real-time voice agents, NLP pipelines, and predictive analytics. Skilled in system design, API development, cloud deployment, and integrating advanced AI models (OpenAI, Anthropic) into user-facing products. Passionate about leveraging artificial intelligence to solve complex real-world problems across diverse sectors like healthcare, recruitment, and agriculture.

PROFESSIONAL EXPERIENCE

Software Engineer

October 2025 – Present

Colossalhub

- Engineered and deployed an end-to-end AI recruitment system, automating CV parsing, candidate shortlisting, and interview workflows to reduce manual effort by 60-70%.
- Built a real-time AI interview agent using LiveKit, capable of conducting voice-based interviews with transcription, evaluation, and session recording capabilities.
- Designed a multi-agent AI pipeline for candidate analysis, shortlisting, and bias detection, significantly improving hiring decision quality and transparency.
- Implemented asynchronous CV processing pipelines, enhancing system responsiveness and user experience under high load conditions.
- Integrated LLM APIs (OpenAI, Anthropic) at scale, managing dynamic token usage while maintaining cost-efficiency and system stability.
- Managed DevOps and infrastructure, including Nginx, Docker, Supervisor, and Linux server environments, ensuring high system reliability and efficient deployments.

Full-Stack Developer Intern

November 2024 – April 2025

Vanguard Economics Ltd Rwanda

- Built and maintained 3 enterprise dashboards for Finance, HR, and Business Development, significantly improving data visibility and decision-making across departments.
- Developed and managed over 15 RESTful API endpoints using Django REST Framework and Node.js, enabling seamless integration between internal and external systems.
- Improved system performance by 30-40% through strategic implementation of Redis caching and comprehensive backend optimizations.
- Reduced hosting and infrastructure costs by approximately 25% by consolidating multiple backend systems into a unified, efficient architecture.
- Successfully integrated external APIs, including Seamless HR and various payment systems, enhancing operational automation and efficiency.
- Achieved 2nd place in an AI hackathon, demonstrating strong problem-solving skills and innovative application of artificial intelligence.

Software Engineer Intern

July 2024 – November 2024

SpiderBit, Rwanda

- Delivered 3+ fully functional web applications, including a blog system and task management platform, significantly enhancing hands-on full-stack development capabilities.
- Built an inventory management system handling over 100 simulated records, featuring automated stock balancing and comprehensive reporting functionalities.
- Developed and deployed a Community Health Reporting Platform backend using Django, supporting structured communication workflows and improving data organization efficiency.
- Designed and documented over 10 RESTful API endpoints, improving system integration and maintainability across various applications.
- Implemented an initial machine learning model with a baseline accuracy of approximately 20%, laying foundational groundwork for future predictive health analytics.
- Reduced manual data handling by automating core inventory and reporting processes, streamlining operations and improving data accuracy.

EDUCATION

Bachelor of Technology Honours Degree in Software Engineering

2026

Harare Institute of Technology

A-Level Subjects: Pure Mathematics, Physics, Chemistry, Computer Science

KEY SKILLS

Programming Languages & Frameworks: Python • TypeScript • React • Next.js • Django • Node.js • Frappe/ERPNext

AI & Data: AI/LLM Integration (OpenAI, Anthropic) • NLP Pipelines • Predictive Analytics • Machine Learning • MongoDB

DevOps & Infrastructure: Docker • Nginx • Supervisor • systemd • Redis • REST APIs • LiveKit

Professional & Soft Skills: System Design • Problem-Solving • Collaboration • Technical Documentation • Agile Methodologies

REFERENCES

Available upon request